Harder Geometrical Problems — problem 12.3.2.84

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A sphere of radius r is inscribed in a tetrahedron. Tangent planes parallel to the faces of the tetrahedron cut off four small tetrahedra from the tetrahedron having inscribed spheres of radii r_1, r_2, r_3, r_4. Then $r_1 + r_2 + r_3 + r_4 = 2r$.